

Title

Study protocol: investigating subanesthetic esketamine as an adjunctive analgesic in emergency laparoscopic surgery for acute gynaecological conditions

Abstract

Background: Postoperative pain hyperalgesia and central sensitization contribute to both acute severe pain and long-term chronic pain sequelae. Novel analgesic strategies targeting pain pathophysiology are needed. This protocol describes a randomized investigation of perioperative sub-dissociative esketamine combined with patient-controlled analgesia. **Methods:** Patients with ectopic pregnancy requiring urgent laparoscopic salpingectomy (target n=84) will be randomized 1:1 to receive intravenous esketamine bolus (0.25 mg/kg) intraoperatively plus postoperative patient-controlled esketamine-sufentanil analgesia versus enhanced opioid analgesia (higher sufentanil dose with saline placebo). **Primary outcome:** pain sensitivity assessed via standardized questionnaire on postoperative days 1, 2, and 7. **Secondary outcomes:** pain intensity ratings, depressive symptom screening, serum inflammatory biomarkers (BDNF, prostaglandins), adverse events. Sample size calculated for 80% power to detect a 1.2-unit difference in primary pain sensitivity measure. Blinding, randomization, and allocation concealment will ensure methodologic rigor.